Chapter 6 - Conditional Expressions

Some times we want to play Pubg on our phone if the day is sunday.
Sometimes we order Icecream online if the day is sunny.
Sometimes we go hiking if our parents allow.

All these are decisions which depends on if a condition being met.

In python programming too, we must be able to execute ~~infra~~ instructions on a ~~conditions on a computer~~ condition (s) being met. This is what Conditionals are for!

If else and elif in Python.
If else and elif. statements are a multiway decision taken by our program due to certain conditions in our code.

Syntax:

```
if (condition1) :          => if condition 1 is true
        print ("Yes")
elif (condition 2) :       => if condition 2 is true
        print ("no")
else :                     => Other wise
    print ("maybe")
```

Code example :

```
a = 22
if (a > 9):
    print ("greater")
else:
    print (" smaller")
```

Quick Quiz: Write a program to print yes when the age entered by the user is greater than or equal to 18.

Sol :

```
~~age = 18~~
~~if (age~~
age = ~~user~~ int(input ("enter your age :"))
if (age >= 18) :
    print (" yes")
elif (age < 18) :
    print (" no")
else :
    print ("not valid") .
```

Relational Operators :
Relational operators are used to evaluate conditions inside the if statements. some examples of relational operators are :

= = → equals
> = → greater than/equal to
<= → less then/equal to etc.

logical operators :
In python logical operators operate on conditional statements Example :

and → true if both operands are true else false.
or → true if at least one operand is true else false.
not → inverts true to false and false to true.

elif clause
elif in python means [else if]. An if statement can be chained together with a lot of these elif statements followed by an else statement

```
if (condition 1):
    # code
elif (condition 2):
    # code
elif (condition 3):
    # code
else :
    # code
```

⇒ This ladder will stop once a condition in an if or elif is met.

Important notes :
1> There can be any number of elif statements.

2> last else is executed only if all the conditions inside elif fails.